GHULAM AHMAD

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SUMMARY

Full Stack AI/ML Engineer building scalable AI-powered systems with LLMs, FastAPI/Node.js, MERN/Next.js/TypeScript. Skilled in LangChain/LangGraph, RAG, applied ML, and no/low-code.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, C++
- **AI / ML:** TensorFlow, Pandas, NumPy
- **Generative AI & LLM Systems:** LangChain, LangGraph, RAG Pipelines, LLM Application Development, AI Agents
- **Frontend:** React.js, Next.js, Tailwind CSS
- **Backend:** Node.js, Express.js, FastAPI
- **Databases & Vector Stores:** MongoDB, SQL Server, Pinecone
- **Real-Time & Distributed Systems:** Socket.IO, RabbitMQ, Redis, LiveKit, Microservices, Event-Driven Systems
- **Cloud & Developer Tools:** MongoDB Atlas, Vercel, Git, GitHub, Nodemon
- **Automation & AI Workflows:** n8n, Workflow Automation

EXPERIENCE

DevTech Fusion ***Sep 2025 – Present***

Part-Time Full Stack AI Engineer

- Designed and developed **8+ AI systems** across customer support, legal tech, hiring automation, and analytics platforms using **LLM-first architectures** such as RAG pipelines, multi-agent workflows, and automated decision engines.
- Built scalable **AI backends using FastAPI, Node.js, LangChain, and vector databases**, powering intelligent applications that reduced manual processes by **70–90%**.
- Developed **real-time AI products** including a sub-second latency voice sales agent, AI-powered customer support chatbot, and an **NL→SQL analytics assistant** achieving **90%+ query accuracy across 80+ database tables**.
- Contributed across the **full stack** by integrating **Next.js/TypeScript frontends with AI backends**, delivering platforms used in real business workflows.

PROJECTS

Umeed: AI Fraud Detection Crowdfunding Platform | *Next.js, Express.js, MongoDB, Flask, Stripe API, XGBoost*

- Built a **bilingual (English/Urdu) crowdfunding platform** with Next.js Server Actions and Stripe PaymentIntent integration, enabling **direct creator payouts** and **real-time donation tracking across 100+ transactions** via dynamic user profiles and dashboards.Trained an XGBoost model to detect fraudulent transactions, achieving 91% accuracy
- Implemented **NextAuth authentication, optimized MongoDB data layer,** and **secure Stripe payment workflows**, supporting **PKR transactions** and **reducing backend boilerplate by ~70%** using a server-action architecture.

DraftFlow: AI-Powered Contract Analysis Platform | *FastAPI · LangChain · OpenRouter · Pinecone · Redis. OCR*

- **RAG-based contract intelligence system** enabling semantic search and conversational querying across **multi-contract datasets** using Pinecone (1536-dim embeddings).
- Implemented **LLM intent routing, multi-contract reference resolution, OCR ingestion,** and **context optimization**, reducing token usage by **~40%** and enabling contract generation, modification, and analysis workflows.

Accountech: AI Invoice Management System | Next.js 15 · TypeScript · React · Tailwind CSS · REST APIs

- Built an **AI invoice processing pipeline** using GPT-4 + Google Vision OCR with **95%+ extraction accuracy**, automating document classification, structured field extraction, and fuzzy client matching to **reduce manual data entry by ~80%**.
- Engineered **parallel batch processing (10 workers)** with async FastAPI services, achieving **6–10× faster processing**, with automated validation (duplicate detection, tax correction) and **intelligent categorization using 158+ multilingual keywords**.

Resume Screening & Hiring Automation | n8n · GPT-4o · Gmail API · Google Drive · Google Sheets · OCR

- Built an **AI-powered resume screening pipeline** using n8n, GPT-4o, and Gmail ingestion with OCR + multi-format parsing, **automating 90% of resume screening** and generating structured candidate scoring outputs via Google Drive and Google Sheets.

Personalized Farmer Harvest Video Automation| n8n · HeyGen · Google APIs · WhatsApp API

- Built a **production-grade AI video automation pipeline** generating and delivering **1K+ personalized videos/day**, implementing batch scheduling, rate limiting, retries, cloud storage, and WhatsApp delivery with **automated tracking that reduced manual workload by ~90%**.

Real-Time AI Voice Sales Agent | LiveKit · Cerebras GPT-OSS 120B · Cartesia STT/TTS · WebRTC

- Built a real-time AI voice sales agent using LiveKit + Cerebras GPT-OSS 120B enabling sub-second latency conversations with full-duplex STT/TTS and **95%+ speech recognition accuracy** for natural customer interactions.
- Designed a **multi-agent system (Sales, Technical, Pricing)** with intelligent handoffs, Silero VAD turn-taking, and dynamic document context injection, **reducing hallucinations by ~80%** and **enabling scalable voice-based sales automation**.

Ping: Real-Time Chat App | *TypeScript, MERN, Socket.IO, RabbitMQ, Redis, TensorFlow*

- Built a **3-microservice real-time chat system** supporting **1K+ concurrent socket connections** with Socket.IO messaging, presence tracking, typing indicators, and read receipts.
- Implemented **passwordless OTP authentication** using Redis (5-min TTL) and RabbitMQ async email delivery, **reducing auth latency ~60%** via event-driven architecture.

POSpal: POS NL→SQL AI Chatbot | Python· DeepSeek v3.1 (OpenRouter) · MySQL·MERN

- **Built an NL→SQL AI chatbot for an 80+ table POS database, achieving 90%+ query accuracy using DeepSeek v3.1 with schema-aware prompt engineering** and **conversational response generation.**
- Engineered a **schema relevance** and **summarization pipeline** (fuzzy matching + FK expansion) reducing **LLM schema context by ~85%** and **token usage by ~95%**, enabling faster and cost-efficient retail data querying.

Real-Time AI Customer Assistant | LangChain · ChromaDB · FastAPI ·MERN·Python · Next.js·

- Built a RAG chatbot for a company website **ingesting 100+ document pages with PDF parsing, chunking,** and **embeddings, enabling instant context-aware answers** and **replacing static FAQ workflows.**
- Engineered a **FastAPI + LangChain retrieval pipeline with persistent ChromaDB storage** and **SSE streaming**, delivering **real-time token-by-token responses** and **reducing manual customer query handling by ~80%**.

EDUCATION

Lahore Garrison University, Lahore *Sep 2023 – May 2027*

Bachelor of Computer Science ***(CGPA***: 3.3/4)

CERTIFICATIONS

- Machine Learning Specialization – Coursera, Stanford Online , Deeplearning.AI
- Deep Learning Specialization – Coursera, Stanford Online , Deeplearning.AI